

Lab assignment- 2.5

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Task -1: Refactoring Odd/Even Logic (List Version)

Prompt : Write a Python program to calculate the sum of odd and even numbers in a list

Code and output :

Explanation

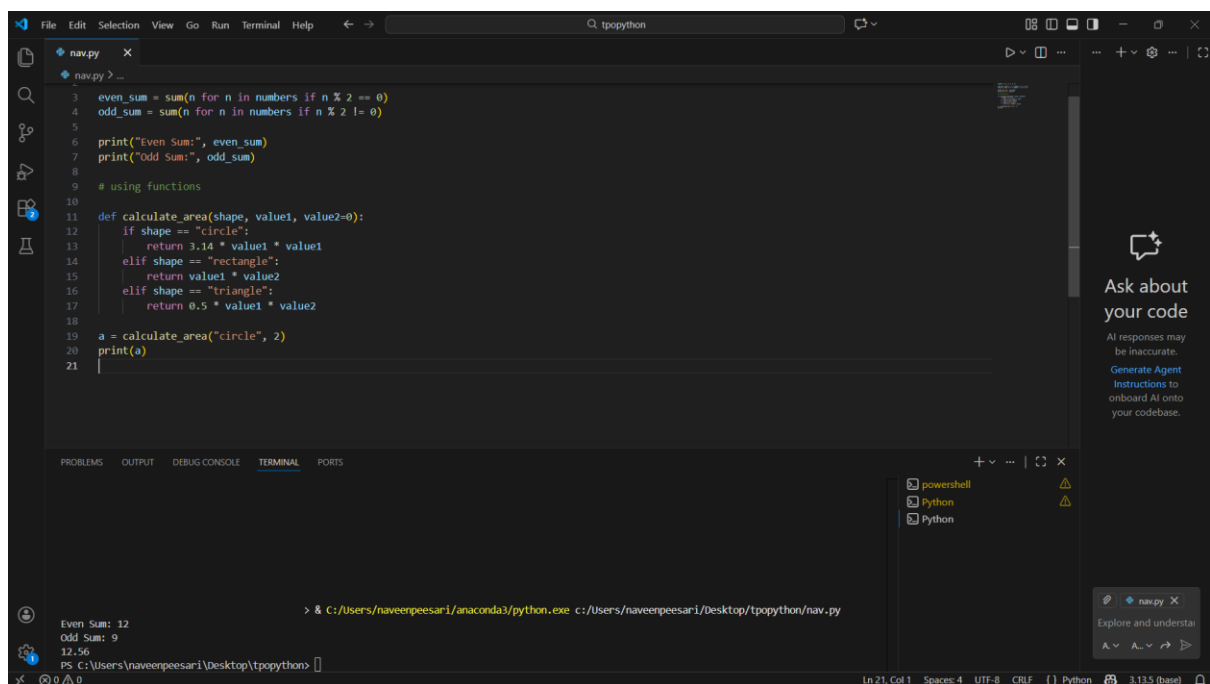
The refactored code is shorter, more readable, and efficient.

It removes manual loops and uses Python's built-in `sum()` with conditions, making the code easier to maintain.

Task 2: Area Calculation Explanation

Prompt : Explain a Python function that calculates the area of different shapes

Code and output :



```
File Edit Selection View Go Run Terminal Help
nav.py
3 even_sum = sum(n for n in numbers if n % 2 == 0)
4 odd_sum = sum(n for n in numbers if n % 2 != 0)
5
6 print("Even Sum:", even_sum)
7 print("Odd Sum:", odd_sum)
8
9 # using functions
10
11 def calculate_area(shape, value1, value2=0):
12     if shape == "circle":
13         return 3.14 * value1 * value1
14     elif shape == "rectangle":
15         return value1 * value2
16     elif shape == "triangle":
17         return 0.5 * value1 * value2
18
19 a = calculate_area("circle", 2)
20 print(a)
21
```

Even Sum: 12
Odd Sum: 9
12.56
PS C:\Users\naveenpeesari\Desktop\tpython>

Explanation

Gemini clearly explains how the function works for different shapes.

It describes the parameters, logic flow, and formulas used, which helps beginners understand the code easily.

Task 3: Prompt Sensitivity Experiment

Prompt 1: Write a Python program to calculate the sum of even and odd numbers in a list

```
1 # Program to calculate the sum of even and odd numbers in a list
2
3 def sum_even_odd(numbers):
4     """Calculate the sum of even and odd numbers in a list"""
5     sum_even = 0
6     sum_odd = 0
7
8     for num in numbers:
9         if num % 2 == 0:
10             sum_even += num
11         else:
12             sum_odd += num
13
14     return sum_even, sum_odd
15
16
17 # Main program
18 if __name__ == "__main__":
19     # Get list of numbers from user
20     user_input = input("Enter numbers separated by spaces: ")
21     numbers = list(map(int, user_input.split()))
22
23     # Calculate sums
24     even_sum, odd_sum = sum_even_odd(numbers)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Unexpected token '14' in expression or statement.
At line:1 char:16
+ 10 11 12 13 14 15
+ ~~~~~
Unexpected token '15' in expression or statement.
+ CategoryInfo : ParserError: (:) [], ParentContainsErrorRecordException
+ FullyQualifiedErrorId : UnexpectedToken

PS C:\Users\NaveenPeesari\Desktop\tpython> C:\Users\NaveenPeesari\anaconda3\python.exe c:/Users/naveenpeesari/Desktop/tpython/nav.py
Enter numbers separated by spaces: 1 2 3 4 5 6 7
Sum of even numbers: 12
Sum of odd numbers: 16
PS C:\Users\NaveenPeesari\Desktop\tpython>

Explanation:

For **Prompt 1 (Basic Prompt)**, Cursor AI generated a simple loop-based program using conditional statements. This version is easy to understand and suitable for beginners, but it uses more lines of code and manual variable updates.

Prompt 2: Write a clean and readable Python program to find the sum of even and odd numbers in a list suitable for beginners

Code and output:

Explanation : For **Prompt 2 (Readability-Focused Prompt)**, the AI produced code with clearer variable names and better formatting. Although the logic is similar to the basic version, readability and clarity were improved, making the code easier to review and maintain.

Prompt 3: Write an optimized Python program to calculate the sum of even and odd numbers in a list using built-in functions

Code and output:

```
1 numbers = (1, 2, 3, 4, 5, 6)
2
3 even_sum = sum(n for n in numbers if n % 2 == 0)
4 odd_sum = sum(n for n in numbers if n % 2 != 0)
5
6 print("Even Sum:", even_sum)
7 print("Odd Sum:", odd_sum)
8
```

Even Sum: 12
Odd Sum: 9
PS C:\Users\NaveenPeesari\Desktop\tpython>

Explanation: For **Prompt 3 (Optimized Prompt)**, Cursor AI generated a more efficient solution using Python's built-in `sum()` function along with conditions. This version reduced the number of lines and improved code efficiency while maintaining correctness.

Prompt 4 : Write a Python program to calculate the sum of even and odd numbers in a list using functions

Code and output :

```
1 def calculate_even_odd_sum(numbers):
2     even_sum = sum(n for n in numbers if n % 2 == 0)
3     odd_sum = sum(n for n in numbers if n % 2 != 0)
4     return even_sum, odd_sum
5
6
7 nums = [1, 2, 3, 4, 5, 6]
8
9 even, odd = calculate_even_odd_sum(nums)
10
11 print("Even Sum:", even)
12 print("Odd Sum:", odd)
13
```

Even Sum: 12
Odd Sum: 9
PS C:\Users\NaveenPeesari\Desktop\tpython>

Explanation:

For **Prompt 4 (Function-Based Prompt)**, the AI created a modular solution using a user-defined function. This approach improves reusability, debugging ease, and maintainability, making it suitable for larger applications.

Task 4: Tool Comparison Reflection

Reflection

Based on the experiments performed in this lab, Google Gemini, GitHub Copilot, and Cursor AI each have different strengths.

Google Gemini is very useful for understanding code, as it provides clear explanations and works well in Google Colab, especially for beginners.

GitHub Copilot offers real-time code suggestions inside VS Code and is best suited for daily development and writing production-ready code.

Cursor AI is effective for experimenting with different prompts, refactoring code, and analyzing multiple coding approaches.